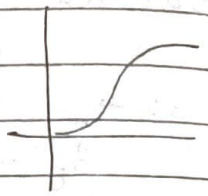
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Lecture - 31
Logistic Regression & GLMs

- Reg. $Y \rightarrow$ quantitative $\quad E[Y|X]$
  Classificat^n $Y \rightarrow$ qualitative $\quad P(Y=k|X)$
- Coding categories as nos. smuggles in an ordering & a spacing. Neither is usually justified.
- Linear model fails on probab.
  ↳ we need a func^n that is linear in the params. but output is confined to $(0,1)$.
- Logistic Model

$$P(Y=1|X=x) = \frac{1}{1+e^{-(\beta_0+\beta_1 x)}}$$



or $\quad \underbrace{\log\left(\frac{P(Y=1|X)}{P(Y=0|X)}\right)}_{\text{log-odds or logit.}} = \beta_0+\beta_1 x$

- $L_n(\beta_0,\beta_1) = \prod_{i=1}^{n} \left(\frac{e^{\beta_0+\beta_1 x_i}}{1+e^{\beta_0+\beta_1 x_i}}\right)^{x_i} \left(\frac{1}{1+e^{\beta_0+\beta_1 x_i}}\right)^{1-x_i}$

$$\beta^{new} = \beta^{old} - \left(\nabla^2 \ell(\beta^{old})\right)^{-1} \nabla \ell(\beta^{old})$$

- Sensitivity (Recall) $= \frac{TP}{TP+FN}$

  Precision $= \frac{TP}{TP+FP}$

  Specificity $= \frac{TN}{TN+FP}$

- regularized log. reg.

$$\min_w \; -\ell(w) + \lambda \|w\|_2^2$$

or

$$\min_w \; -\ell(w) + \lambda \|w\|_1$$

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it helps in controlling complexity, penalizes large coefficients & prevents overfitting. L2 shrinks coefficients smoothly & improves numerical stability & promotes sparsity.

- Regularizn also rescues the fit under perfect separa", where the unpenalized MLE diverges.
- If Y is a count → OLS won't work as predicⁿ can be -ve, var. is not constant & fitted values must be bounded.

$Y_i \sim \text{Poisson}(\mu_i)$

$\log(\mu_i) = x_i^T \beta$

$\Rightarrow \quad \mu_i = \exp(x_i^T \beta) > 0$

mean = variance, varies with i & no negative predictions

- GLM has 2 things —

1. response from the exp. family

$$f(y|\theta, \phi) = \exp\left[\frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi)\right]$$

$\theta \rightarrow$ canonical param. (locaⁿ)

$\phi \rightarrow$ dispersion (scale)

for gaussian ϕ is free, for binomial & poisson it is 1.

Mean depends on θ

$\text{Var}(\mu) = b''(\theta)\, a(\phi)$

2. link funcⁿ related to linear predictor

$$\eta = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p = x^T \beta$$

$$\eta = g(\mu) = \theta$$

$$\hat{\text{Var}}(\hat{\beta}) = (X^T W X)^{-1} \phi$$

↳ diagonal matrix of wts.

- fitting is newton-raphson & each step is a weighted least squares. The likelihood is concave with a unique maximum but there is no closed form because the Hessian depends on β.
- With J categories, every predictor costs J-1 params, & the coefficients describe contrasts against a baseline, not the probab. of any single category.